

Joshua Keene, PhD

(810) 287-5699 | Lansing, MI | keenejosh22@gmail.com | keenejosh.github.io

PROFESSIONAL SUMMARY

Neuroscience PhD specializing in neuroinflammation, tauopathy, and Alzheimer's disease pathology using transgenic and knock-in mouse models; demonstrated p-tau (AT8) elevations of 5.4-fold in hippocampus and 4.7-fold in cortex persisting 30 days post-systemic PAMP challenges in PS19 mice and a 52% paradoxical reduction in hippocampal amyloid burden following prenatal inflammatory stress in 5xFAD mice. First-author manuscript describing PAMP-induced tau pathology and neuroinflammation currently in final preparation. Research experience spans in vivo preclinical studies, primary cell culture and in vitro assays, quantitative histology, advanced imaging, and biostatistical analysis in R. Prior clinical neuropsychology experience provides direct exposure to neurocognitive phenotypes across TBI, dementia, and neurodevelopmental populations.

EDUCATION

2018–2024 PhD in Neuroscience

Central Michigan University, Mount Pleasant, MI | Advisor: Dr. Gary Dunbar

Dissertation: *PAMP Exposure Increases Tau and Inflammation in Alzheimer's Disease.*

2015–2018 MS in Experimental Psychology

Central Michigan University, Mount Pleasant, MI | Advisor: Dr. Yannick Marchalant

Master's Thesis: *Role of Early Exposure to Inflammatory Stress on Plaque Deposition and Microglia Density in Alzheimer's Disease Using the 5xFAD Mouse Model.*

2010–2013 BA in Psychology, Minor in Philosophy

Michigan State University, East Lansing, MI

RESEARCH INTERESTS

Neuroinflammation | Aging | Alzheimer's Disease | Neurodegeneration | Tauopathy | Amyloid Pathology | Innate Immune Signaling | Glial Biology | Peripheral Immune Activation | Preclinical Mouse Models

MANUSCRIPTS IN PREPARATION FOR SUBMISSION

In Prep

Keene, J., Desjarlais, T., Jackman, B., Suresh, S., Rossignol, J., Gordon, M., Dunbar, G.L., & Morgan, D. Multiple Systemic PAMP Administrations Increase Long-Lasting Tau Phosphorylation and Neuroinflammation in PS19 Tauopathy Mice. First-author manuscript in final preparation for submission.

JOURNAL PUBLICATIONS

- 2015** Rohling, M. L., Meyers, J. E., Williams, G. R., Kalat, S. S., Williams, S. K., & Keene, J. (2015). Application of the Daubert Standards to the Meyers Neuropsychological Battery Using the Rohling Interpretive Method. *Psychological Injury and Law*, 8(3), 252-264.

PROFESSIONAL EXPERIENCE

2022–2024 Collaborative Graduate Researcher

Laboratory of Dr. David Morgan and Dr. Marcia Gordon | Michigan State University, Grand Rapids Research Center (GRRC), Grand Rapids, MI

- Served as primary cross-institutional researcher (CMU/MSU-GRRC) on a multi-year preclinical study examining whether repeated systemic administration of multiple PAMPs (Zymosan A, Poly I:C, ODN 2395, LPS; TLR2/3/4/9 agonists) drives persistent tau pathology and neuroinflammation in PS19 tauopathy mice; first-author manuscript in final preparation for submission.
- Demonstrated persistent p-tau (AT8) elevations of 5.4-fold in hippocampus and 4.7-fold in cortex persisting 30 days post-final PAMP challenge in PS19 mice, with total human tau (HT7) remaining unaltered, confirming post-translational hyperphosphorylation at pS202/pT205 without corresponding changes in transgene expression.
- Characterized strong positive Spearman correlations between p-tau burden and glial reactivity markers in the hippocampus (MHC II: $r_s=0.523$; GFAP: $r_s=0.556$; CD68: $r_s=0.501$), suggesting a close bidirectional relationship between p-tau accumulation and glial activation following peripheral immune challenges.
- Performed microtome sectioning and DAB IHC on hemi-brain tissue across a comprehensive antibody panel targeting tau pathology (HT7, AT8), neuroinflammation (GFAP, MHC II, CD68, Iba1, CD45, CD11b, CD11c); conducted sandwich ELISA on fresh-dissected hippocampal and cortical tissue; acquired whole-slide digital images via ZEISS Axio Scan.Z1 and quantified percent immunopositive area in NearCYTE with marker-specific standardized thresholds applied consistently across all experimental groups; scored and analyzed NOR and SOR behavioral data using ANY-maze software.
- Conducted parallel 3xTg cohort arm examining whether amyloid burden modulates PAMP-induced tau pathology and neuroinflammation in a combined amyloid-tau model relative to PS19; identified absence of expected amyloid pathology through 6E10, Congo Red, and H31L21 validation staining, confirming phenotypic divergence consistent with genetic drift, which drove the strategic discontinuation of the model arm.
- Conducted all biostatistical analysis in R and GraphPad Prism; applied non-parametric methods throughout (majority of measures non-normally

distributed: 14/17 Shapiro-Wilk/Anderson-Darling; heteroscedastic: 8/17 Brown-Forsythe), including ART-ANOVA for genotype x treatment interactions, Mann-Whitney U for PS19-specific tau markers, and Spearman correlation for glial-tau relationships; reported bootstrapped effect sizes (Cliff's Delta, Hodges-Lehmann; 5,000-10,000 resamples) with Holm-corrected post-hoc contrasts.

2020–2024 Doctoral Researcher

Laboratory of Dr. Gary Dunbar | Central Michigan University, Mount Pleasant, MI

- Conducted doctoral dissertation research investigating whether repeated systemic PAMP administration drives persistent tau pathology and neuroinflammation in PS19 tauopathy mice; managed all aspects of a 150+ animal pipeline across PS19 and 3xTg cohorts including colony maintenance, breeding, and genotyping (tail snip, PCR); doctoral dissertation completed in 2024.
- Executed all in vivo experimental procedures including PAMP preparation and concentration dilutions, monthly IP injections over a four-month period, systematic post-injection monitoring of sickness behavior (hunched posture, hypoactivity, piloerection) with supportive care (mash diet) until recovery, and behavioral testing (hindlimb clasp, open field (HK MotorMonitor), Morris water maze, NOR, SOR).
- Conducted hemi-brain collections at a 30-day post-final-injection endpoint to capture long-lasting rather than acute inflammatory effects, including fresh tissue microdissections of hippocampus and cortex and PFA-fixed hemisphere for histology; tissue subsequently processed and analyzed at MSU-GRRC.
- Supervised and trained undergraduate researchers in behavioral testing, animal handling, and tissue processing procedures.

2018–2020 Graduate Researcher

Laboratory of Dr. Kevin Park | Central Michigan University, Mount Pleasant, MI

- Independently executed two concurrent studies (120+ animals) in the TAg/OFF bitransgenic neuronal cell-cycle re-entry (NCCR) model of sporadic AD: (1) TAg/OFF/MAPT triple transgenic investigation of whether NCCR combined with humanized tau (KI) drives AD-like tauopathy and neuroinflammation, and (2) systemic PAMP administration study examining whether peripheral immune activation exacerbates neuroinflammation in NCCR mice; validated model specificity through single transgene controls (TAg-only, OFF-only) confirming pathology required bitransgenic co-expression.
- Identified moderate p-tau accumulation (CP13) in cortical and hippocampal neurons of TAg/OFF/MAPT mice undergoing NCCR; characterized hippocampal and cortical neuroinflammation via multi-label IF co-staining (Iba1/CD45/Hoechst, Iba1/GFAP/Hoechst) demonstrating reactive microglial and astrocytic profiles in NCCR animals; observed elevated glial reactivity in PAMP-treated animals relative to vehicle controls.
- Executed all experimental procedures across both studies including colony maintenance and breeding, genotyping (tail snip, multiplex PCR), behavioral

testing (open field, EPM, Y-maze, NOR), transcardial perfusions (isoflurane, 4% PFA), fresh tissue microdissections (hippocampus, cortex), vibratome sectioning, formic acid antigen retrieval, and multi-label IF and DAB IHC across a comprehensive panel (15+ targets) including tauopathy markers (ALZ50, MC-1, CP13, CP27, RZ3, DA9, TOC1, PHF-1), cell cycle markers (SV40T, PCNA), neuroinflammatory markers (Iba1, CD45, GFAP, MHC II, CD68), and neuronal integrity markers (NeuN, NeuroTrace/Nissl); acquired whole-section digital images using ZEISS Axio Imager M1.

- Co-managed PLX5622 CSF1R-mediated microglial knockdown study in TAg/OFF mice; confirmed pronounced Iba1-positive microglial knockdown via immunofluorescence imaging; contributed colony maintenance, genotyping, transcardial perfusions, microdissection, vibratome sectioning, and IHC support to Barrett et al. APP-NL/NL-F x TAg/OFF amyloid-NCCR study; presented findings at MADRC, MPS, and SCREE (2019).

2018

Visiting Graduate Researcher

Laboratory of Dr. David Morgan and Dr. Marcia Gordon | Michigan State University, Translational Science and Molecular Medicine (TSMM), Grand Rapids, MI

- Performed AAV-mediated stereotaxic injections of human tau constructs into bilateral hippocampus and anterior cortex (4 injection sites per animal, 25+ animals) for a tau seeding and propagation dose-optimization study; prepared viral vector and loaded Hamilton syringes for stereotaxic delivery.
- Conducted systematic post-operative monitoring and recovery scoring over 7 days (hypoactivity, hunched posture, social behavior, wound healing, body weight) with supportive care (mash diet) in TSMM vivarium; performed transcardial perfusions (4% PFA), microtome sectioning, and DAB IHC (HT7, AT8, pSer396).

2015–2018

Graduate Researcher

Laboratory of Dr. Yannick Marchalant, Neuroinflammation, Aging, and Alzheimer's Lab | Central Michigan University, Mount Pleasant, MI

- Conducted independent master's thesis research investigating prenatal inflammatory effects on amyloid burden and microglial density in the 5xFAD transgenic mouse model; managed colony maintenance, breeding, genotyping (tail snip, multiplex PCR), transcardial perfusions (4% PFA), vibratome sectioning, multi-label IF co-staining (6E10, Iba1, Hoechst), widefield (ZEISS Axio Imager M1) and confocal (Nikon A1R) microscopy, and unbiased stereological quantification (MBF Stereo Investigator); executed parallel acute IP LPS arm (1 mg/kg) in 2-month-old 5xFAD mice, collected at 4 months, examining systemic inflammatory effects on amyloid pathology at the prodromal stage; Master's thesis completed in 2018.
- Demonstrated that prenatal inflammatory exposure paradoxically reduced hippocampal amyloid plaque burden by ~52% in 5xFAD offspring of transgenic mothers relative to offspring of wild-type mothers, with significant reductions

observed independently in both sexes; maternal inflammatory exposure did not alter microglial density, suggesting altered microglial function and/or amyloid processing independent of microgliosis, identifying early immune signaling as a modifiable contributor to amyloid pathology.

- Designed and executed dose-response studies examining cannabinoid agonist WIN-55,212-2 effects on MSC proliferation, migration, and wound healing; identified a biphasic proliferative response peaking at 1 μ M relative to vehicle and 10 μ M concentrations across 3- and 7-day time points; validated MSC phenotype via FACS (Sca1+/CD11b-) in collaboration with the Rossignol lab; isolated and maintained primary MSC, NSC, and microglial cultures from bone marrow and neonatal brain tissue; contributed to Bruins et al. cannabinoid-MSC modulation study; presented findings at MISfN (2016 and 2019).
- Independently conducted Dil lipophilic neuronal tracing in thick (200 μ m) 5xFAD vibratome sections to quantify dendritic spine density and arborization across hippocampal and cortical regions; contributed confocal imaging and MBF Neurolucida neuronal reconstruction to Guentert et al. APP NL-G-F dendritic spine morphology and arborization study; presented findings at MISfN and SCREE (2018).
- Contributed histological processing, immunofluorescence staining (TH, GFAP, Hoechst), and imaging of SNpc and striatum to Crombez et al. LRRK2-KO and Altemus et al. LRRK1-KO Parkinson's disease studies examining LRRK paralog roles in dopaminergic degeneration and neuroinflammation, in collaboration with the Johansen lab; presented findings at WMRUGS and mGluRs (2016) and SCREE (2017).
- Conducted cross-sectional study examining age-related changes in dentate gyrus neurogenesis (DCX, NeuN), hippocampal and cortical neuroinflammation (Iba1, GFAP), and ghrelin receptor (GHSR1a) expression in middle-aged to aged Sprague-Dawley rats (14–24 months) in collaboration with the Sharma lab (Exercise Physiology, CMU); presented findings at MISfN, SCREE, and BRAIN Symposium (2016).
- Supervised, trained, and mentored 30+ undergraduate and medical student researchers across 3+ years in transcardial perfusions, vibratome sectioning, immunohistochemistry, cell culture, behavioral testing, microscopy, and statistical analysis; designed and delegated independent research projects to advanced trainees, resulting in 6 undergraduate poster presentations at regional conferences.

2013–2015 Lead Psychometrist

Clinical Director Dr. Gerard Williams, Delta Family Clinic South P.C., Flint, MI

- Administered, scored, and synthesized neuropsychological assessment batteries across pediatric and adult populations, including the Meyers Neuropsychological Battery (MNB), Wechsler scales (WPPSI, WISC, WAIS), MMPI-2, PAI, CARS-2, and GADS; evaluated neurocognitive and neurobehavioral profiles across clinical populations, including traumatic brain injury (TBI), stroke, mild cognitive impairment (MCI)/dementia, attention-deficit/hyperactivity disorder (ADHD), and autism spectrum disorder (ASD).

- Trained and supervised fellow psychometrists in standardized test administration and scoring protocols.

2004–2005 IT Support Technician (Co-Op)

Swartz Creek Community Schools, Swartz Creek, MI.

- Contributed to school district security camera system design using CAD software; provided hardware and software troubleshooting and deployment across district facilities.

POSTER PRESENTATIONS

- 2023** **Keene, J.**, Desjarlais, T., Jackman, B., Srinageshwar, B., Finneran, D., Suresh, S., Henry, L., Gordon, M., Rossignol, J., Morgan, D., & Dunbar, G. (2023). “Inflammatory Stress Contributes to Tau Pathology in the PS19 Mouse Model of Alzheimer’s Disease.” Presented at the Student Creative and Research Endeavors Exhibition (SCREE), Central Michigan University (CMU), Mount Pleasant, MI.
- 2022** **Keene, J.**, Srinageshwar, B., Suresh, S., Stanley, R., Rossignol, J., & Dunbar, G. (2022). “Peripheral Inflammation and Physiological Hallmarks of Alzheimer’s Disease.” Presented at SCREE, CMU, Mount Pleasant, MI.
- 2019** Barrett, T., **Keene, J.**, Saito, T., Saido, T., & Park, K.H. (2019). “SV40T-Mediated Neuronal Dysregulation Induces NFT Pathology in APPNLF/NLFI Mice.” Presented at the Michigan Alzheimer’s Disease Research Center (MADRC) 3rd Annual Research Symposium, Grand Rapids, MI and Michigan Physiological Society 6th Annual Meeting, Mount Pleasant, MI.
- Keene, J.**, Madsen, L., Ginster, L., Bruins, A., Sawaya III, G., Barrett, T., Guentert, L., & Marchalant, Y. (2019). “Effects of Inflammatory Stress During the Asymptomatic Phase of Alzheimer’s Disease.” Presented at the Michigan Society for Neuroscience (MISfN) Annual Meeting, Michigan State University (MSU), East Lansing, MI.
- Bruins, A., **Keene, J.**, & Marchalant, Y. (2019). “Synthetic Cannabinoid Agonist WIN 55,212-2 Modulates Proliferation and Migration of Bone Derived Murine Mesenchymal Stem Cells in Vitro.” Presented at the MISfN Annual Meeting, MSU, East Lansing, MI.
- Madsen, L., Guentert, L., **Keene, J.**, Dvorak, R., Shimer, F., Bruins, A., Sawaya III, G., & Marchalant, Y. (2019). “The Role of Inflammation and Sex on the Onset and Progression of Alzheimer’s Disease.” Presented at SCREE, CMU, Mount Pleasant, MI.
- MacLean, S., Barrett, T., Gallien, J., **Keene, J.**, Brown, R., McCloskey, K., & Park, K. (2019). “The Effect of CSF1R Antagonist-mediated Ablation of Microglia in TAG/OFF Mice.” Presented at SCREE, CMU, Mount Pleasant, MI.
- 2018** **Keene, J.**, Madsen, L., Ginster, L., Kunz, A., Bruins, A., Sawaya III, G., Guentert, L., Ridgell, R., & Marchalant, Y. (2018). “Role of Early Exposure to Inflammatory Stress on Plaque Deposition and Microglia Density in Alzheimer’s Disease.” Presented at

the MISfN Annual Meeting, Wayne State University (WSU), Detroit, MI and SCREE, CMU, Mount Pleasant, MI.

Keene, J., Madsen, L., Ginster, L., Kunz, A., Bruins, A., Sawaya III, G., & Marchalant, Y. (2018). "Role of Prenatal Exposure to Inflammatory Stress on Microglia Density and Amyloidogenesis." Presented at SCREE, CMU, Mount Pleasant, MI.

Madsen, L., **Keene, J.,** Guentert, L., Bruins, A., Kunz, A., Ginster, L., Sawaya III, G., & Marchalant, Y. (2018). "Role of Inflammation and Sex in the Onset and Progression of Alzheimer's Disease." Presented at the MISfN Annual Meeting, WSU, Detroit, MI and SCREE, CMU, Mount Pleasant, MI.

Guentert, L., Madsen, L., **Keene, J.,** & Marchalant, Y. (2018). "The Role of Inflammation and Sex on Dendritic Spines and Neuron Arborization in NL-G-F Mice." Presented at the MISfN Annual Meeting, WSU, Detroit, MI and SCREE, CMU, Mount Pleasant, MI.

2017

Keene, J., Madsen, L., Pryde, C., Guentert, L., Crombez, E., McCrary, B., Cersosimo, L., Bailey, P., Kunz, A., Schwab, M., Gutierrez, N., Kemmerer, A., & Marchalant, Y. (2017). "Role of Neuroinflammation and Sex Differences in Alzheimer's Disease." Presented at the MISfN Annual Meeting, University of Michigan (U-M), Ann Arbor, MI and SCREE, CMU, Mount Pleasant, MI.

Madsen, L., **Keene, J.,** McCrary, B., & Marchalant, Y. (2017). "Role of Inflammation and Sex on the Onset and Progression of Alzheimer's Disease." Presented at the MISfN Annual Meeting, U-M, Ann Arbor, MI and SCREE, CMU, Mount Pleasant, MI.

Crombez, E., **Keene, J.,** Altemus, M., Madsen, L., Marchalant, Y., & Johansen, J. (2017). "Influence of Neuroinflammation in the LRRK2 Mouse Model of Parkinson's Disease." Presented at SCREE, CMU, Mount Pleasant, MI.

2016

Crombez, E., **Keene, J.,** Altemus, M., Madsen, L., Johansen, J., & Marchalant, Y. (2016). "Influence of Neuroinflammation in LRRK2 Mouse Model of Parkinson's Disease." Presented at the West Michigan Regional Undergraduate Science (WMRUGS) Research Conference, Grand Rapids, MI and Midwest/Great Lakes Undergraduate Research Symposium (mGluRs) in Neuroscience, Ohio State University, Columbus, Ohio.

Keene, J., & Marchalant, Y. (2016). "Ghrelin's Influence on Neurogenesis and Neuroinflammation in Aged Rats." Presented at the MISfN Annual Meeting, MSU, East Lansing, MI and BRAIN Center Symposium, CMU, Mount Pleasant, MI.

Keene, J., Antcliff, A., Crombez, E., Rossignol, J., & Marchalant, Y. (2016). "Cannabinoid Agonist WIN 55,212-2 Modulation of Bone Derived Mesenchymal Stem Cells." Presented at the MISfN Annual Meeting, MSU, East Lansing, MI.

Keene, J., & Marchalant, Y. (2016). "Ghrelin Induced Hippocampal Neurogenesis and Anti-inflammatory Effects in Aged Rats." Presented at SCREE, CMU, Mount Pleasant, MI.

GRANTS & AWARDS

- 2023** Field Neuroscience Chair Scholarship (\$1,700) | Central Michigan University, Mount Pleasant, MI
- 2019** Steinmetz Graduate Scholarship (\$1,000) | Central Michigan University, Mount Pleasant, MI
- 2018** Graduate Student Research and Creative Endeavors Grant (\$800) | Central Michigan University, Mount Pleasant, MI — “*Effects of Inflammatory Stress During the Asymptomatic Phase of Alzheimer’s Disease.*”
- Steinmetz Graduate Scholarship (\$500) | Central Michigan University, Mount Pleasant, MI
- 2016** Steinmetz Graduate Scholarship (\$1,500) | Central Michigan University, Mount Pleasant, MI
- Graduate Student Research and Creative Endeavors Grant (\$800) | Central Michigan University, Mount Pleasant, MI — “*Ghrelin Induced Neurogenesis and Cell Survival in Aged Rats*”
- 2015–2016** Masters Fellowship in Experimental Psychology | Central Michigan University, Mount Pleasant, MI

TEACHING EXPERIENCE

- 2018** Instructor: *Stress*. Department of Psychology, Central Michigan University, Mount Pleasant, MI.
- Instructor: *Research Methods*, online. Department of Psychology, Central Michigan University, Mount Pleasant, MI.
- Guest Lecturer: “Emotional and Motivated Behavior” and “Sleep and Dreaming” in course *Physiological Psychology*, Department of Psychology, Central Michigan University, Mount Pleasant, MI.
- 2017** Guest Lecturer: “Emotional Behaviors” in course *Behavioral Neuroscience*, Department of Psychology, Central Michigan University, Mount Pleasant, MI.
- Teaching Assistant: *Developmental Psychology*, Department of Psychology, Central Michigan University, Mount Pleasant, MI. — Guest Lecturer: “Middle Adulthood” and “Death and Afterlife Beliefs.”
- Teaching Assistant: *Introduction to Psychology*, Department of Psychology, Central Michigan University, Mount Pleasant, MI. — Guest Lecturer: “Stress and Health.”
- 2016** Guest Lecturer: “Internal Regulation” in course *Behavioral Neuroscience*, Department of Psychology, Central Michigan University, Mount Pleasant, MI.
- Teaching Assistant: *Developmental Psychology*, Department of Psychology, Central Michigan University, Mount Pleasant, MI. — Guest Lecturer: “Late Adulthood.”

PRESENTATIONS, PANELS & WORKSHOPS

- 2021** Presenter: “The Role of Multiple Infection Mimetics on Tau Pathology and Neuroinflammation in PS19 Mice.” Grand Rounds Lecture, Field Neurosciences Institute, Ascension St. Mary’s, Saginaw, MI.
- 2018** Presenter: “Effects of Inflammatory Stress During the Asymptomatic Phase of Alzheimer’s Disease.” Experimental Psychology Seminar, Psychology Department, Central Michigan University, Mount Pleasant, MI.
- Participant: Graduate Student Teaching Preparation Workshop, Psychology Department, Central Michigan University, Mount Pleasant, MI.
- 2017** Panelist: “Graduate Student Perspectives on Neuroscience Research and Graduate School Success.” Neuroscience Night, Nu Rho Psi, Central Michigan University, Mount Pleasant, MI.
- Presenter: “Sex Differences in Alzheimer’s Disease using the 5xFAD Mouse Model.” Experimental Psychology Department, Central Michigan University, Mount Pleasant, MI.
- Presenter: “The Role of Neuroinflammation and Sex Differences in Alzheimer’s Disease.” Experimental Psychology Department, Central Michigan University, Mount Pleasant, MI.
- Panelist: “Graduate School Application and Success Advice.” Psi Chi Undergraduate Panel, Central Michigan University, Mount Pleasant, MI.
- 2016** Presenter: “Ghrelin’s Influence on Neurogenesis and Neuroinflammation in Aged Rats.” Experimental Psychology Department, Central Michigan University, Mount Pleasant, MI.
- Panelist: “Alzheimer’s Disease and Neuroinflammation.” Brain Awareness Conference, Central Michigan University, Mount Pleasant, MI.

PROFESSIONAL SERVICE

- 2024** **Keene, J. & Sanders, G. (Eds.).** (2024). *Foundations of Neuropsychology Vol. 1*. OER Commons / Allan Hancock College, Santa Maria, CA. (Authored by Gallien, J.)
- 2019** **Ad Hoc Reviewer**, Archives of Clinical Neuropsychology (1 manuscript)

OUTREACH & SERVICE

- 2024–2025** Literacy Volunteer, Reading is Fundamental, Pleasant View Magnet School, Lansing School District, Lansing, MI. Classroom reading sessions (Pre-K, Kindergarten); book and educational materials distribution.

ORGANIZATIONS & MEMBERSHIPS

- 2017–2018** Graduate Committee member, Central Michigan University, Mount Pleasant, MI.
- 2017** Graduate Research and Creative Endeavors Grant Reviewer, Central Michigan University, Mount Pleasant, MI.

- 2016–2018** Experimental Psychology Organization Treasurer, Department of Psychology. Central Michigan University, Mount Pleasant, MI.
- 2016–2024** Michigan Society for Neuroscience member.
- 2015–2018** Experimental Psychology Organization member, Central Michigan University, Mount Pleasant, MI.

TECHNICAL SKILLS

Animal Models:

- **Neurodegenerative Disease:** Alzheimer's (transgenic: 3xTg, 5xFAD, PS19; knock-in: APP-KI [NL-F, NL-G-F], MAPT-KI) and Parkinson's (LRRK1/LRRK2 knock-out).
- **Inducible Systems:** Doxycycline-mediated cell cycle re-entry (TAg/OFF NCCR).
- **Viral Vector-Based:** AAV-mediated human tau seeding (bilateral hippocampus and anterior cortex).

Surgical & In Vivo Procedures:

- **Behavioral Testing:** Open field, Morris water maze (MWM), elevated plus maze (EPM), novel object recognition (NOR), spatial object recognition (SOR), Y-maze, hindlimb clasp, behavioral tracking and analysis software (HK MotorMonitor, EthoVision XT, ANY-maze).
- **Surgical Procedures:** Anesthesia (isoflurane, pentobarbital), stereotaxic survival surgery (cannula implantation, osmotic pump implantation, AAV-mediated injection), intraperitoneal (IP) injection, blood collection and serum fractionation, rodent handling and husbandry (mouse/rat), transcardial perfusion (4% PFA), brain microdissection (cortex and hippocampus); pharmacological microglial depletion (PLX5622/CSF1R inhibition).

Histology & Tissue Processing: Tissue sectioning (vibratome, microtome, cryostat), tissue cryopreservation, antigen retrieval (heat-induced/citrate, formic acid, enzymatic/trypsin), immunohistochemistry (DAB, IF, multi-label co-staining), Nissl staining, lipophilic neuronal tracing (DiI).

Microscopy, Stereology & Image Analysis: Confocal microscopy (Nikon A1R), widefield microscopy (brightfield and fluorescence; ZEISS Axio Scan.Z1, ZEISS Axio Imager M1), unbiased stereology (MBF Stereo Investigator: Optical Fractionator, Area Fraction Fractionator, Cavalieri method), dendritic morphology analysis (MBF Neurolucida), image analysis (ImageJ, FIJI, NearCYTE).

Molecular Biology: PCR (nested and multiplex; including primer design), qRT-PCR (multiplex; TaqMan-based gene expression quantification), Western blot, ELISA (sandwich).

Cell Culture & In Vitro: Primary cell culture (NSCs, MSCs, microglia, neurons) and established cell lines (U87 glioblastoma); in vitro assays: drug dose-response, cell proliferation, cell migration, and scratch/wound healing assays; cell passaging and cryopreservation; cell quantification (hemocytometer, manual counting); fluorescence-activated cell sorting (FACS); immunocytochemistry (ICC): Live cell and post-fixed staining of primary cell cultures, targets include CB1, Iba1, Nestin, SOX2, CD73, CD90, CD105; introductory C. elegans techniques (forward genetic screens, RNAi).

Quantitative Analysis & Biostatistics:

- **Software:** R programming, GraphPad Prism, SPSS, GitHub Pages.
- **Methods:** Experimental design, statistical modeling (ANOVA, Mann-Whitney U, Brunner-Munzel, Aligned Rank Transform, Robust Two-Way ANOVA, permutation, Spearman correlation), normality testing (Shapiro-Wilk, Anderson-Darling), post-hoc corrections (Tukey's, Holm); effect size measures (Partial Eta/Omega Squared, Cliff's Delta, Probability Index, Hodges-Lehmann), and data visualization.